Course Title

Machine Learning for Computer Vision: From 2D to 3D Articulated Object Understanding

Course Description

This course introduces core machine learning methods for computer vision, then builds toward advanced topics in **3D scene understanding**, with special focus on **part segmentation** and **articulated object modeling**. Students will learn to represent, process, and reason about 3D data (point clouds, meshes, implicit fields), apply deep learning methods for segmentation and articulation prediction, and implement a semester-long project using state-of-the-art datasets such as **PartNet-Mobility** and **Shape2Motion**.

Course Objectives

By the end of this course, students will be able to:

- Understand machine learning methods for 2D and 3D computer vision.
- Work with 3D data representations and neural architectures (PointNet, Transformers, Implicit Fields).
- Perform 3D **part segmentation** and predict **articulation parameters** (joint type, axis, range of motion).
- Evaluate models on open datasets and extend them to robotic simulation.
- Explore research-level problems like generalization, weak supervision, and sim-to-real transfer.

Prerequisites

- Prior course in machine learning / deep learning.
- Familiarity with Python, PyTorch/TensorFlow.
- Background in linear algebra, probability, and computer vision basics.

Weekly Schedule (14 Weeks)

Weeks 1–2: Foundations

- **Topics**:
    - Intro to ML for CV: supervised, unsupervised, self-supervised learning.
    - From pixels to point clouds: 2D vs 3D vision tasks.
    - Representations of 3D data: voxels, meshes, point clouds, implicit fields.
- **Readings**:
    - PointNet: Deep Learning on Point Sets for 3D Classification and Segmentation (Qi et al., CVPR’17).
    - Survey on Modeling of Human-made Articulated Objects (CGF’25).
- **Lab**: Intro to PyTorch3D / Open3D.
- **Mini-project**: Visualize ShapeNet objects in different representations.

Weeks 3–4: Core 3D ML Architectures

- **Topics**:
    - PointNet++, KPConv, graph-based methods.
    - Transformers for 3D vision.
    - Loss functions: Chamfer, EMD, IoU.
- **Readings**:
    - PointNet++: Deep Hierarchical Feature Learning on Point Sets (Qi et al., NeurIPS’17).
    - Point Transformer (Zhao et al., ICCV’21).
- **Lab**: Implement PointNet for toy segmentation task.

Weeks 5–6: Part Segmentation

- **Topics**:
    - Supervised segmentation on PartNet.
    - Weakly/self-supervised segmentation: contrastive and masked modeling.
- **Readings**:

- PartNet: A Large-Scale Benchmark for Fine-Grained and Hierarchical Part-Level 3D Object Understanding (Mo et al., CVPR'19).
- PointContrast: Unsupervised Pre-training for 3D Point Cloud Understanding (Xie et al., ECCV'20).

- **Lab**: Train a segmentation model on chairs/tables.
- **HW1**: Compare supervised vs. contrastive pretraining for part segmentation.

Weeks 7–8: Articulation Modeling

- **Topics**:
  - Kinematics of articulated objects: revolute vs prismatic joints.
  - Predicting joint type/axis from point clouds.
- **Readings**:
  - Category-Level Articulated Object Pose Estimation (Wang et al., CVPR'20).
  - CAPT: Category-level Articulation Estimation from a Single Point Cloud using Transformers (2024).
- **Lab**: Estimate joint type and axis on Shape2Motion dataset.
- **HW2**: Implement a baseline joint parameter predictor.

Weeks 9–10: Neural Implicit Representations for Articulation

- **Topics**:
  - Signed Distance Functions (SDFs), Occupancy Networks, Neural Radiance Fields.
  - Neural implicit models for articulated objects.
- **Readings**:
  - A-SDF: Learning Disentangled Signed Distance Functions for Articulated Shape Representation (Wu et al., ICCV'21).
  - Neural Implicit Representation for Building Digital Twins of Unknown Articulated Objects (CVPR'24).
- **Lab**: Use an SDF-based network for shape + articulation learning.

Weeks 11–12: Applications in Robotics

- **Topics**:

- Sim-to-real challenges in 3D vision.
  - Using SAPIEN / PyBullet for articulated object interaction.
  - Reinforcement learning for manipulation.
- **Readings**:
  - MARS: Multimodal Active Robotic Sensing for Articulated Objects (IJCAI'24).
  - Online Estimation of Articulated Objects with Factor Graphs (2024).
- **HW3**: Simulate interaction with an articulated cabinet in SAPIEN using predicted joints.

Weeks 13–14: Projects and Future Directions

- **Topics**:
  - Student project presentations.
  - Emerging directions: Gaussian splatting for articulation (ArticulatedGS), open-vocabulary part segmentation, interaction-guided perception.
- **Readings**:
  - ArticulatedGS: Self-Supervised Digital Twin Modeling of Articulated Objects using 3D Gaussian Splatting (CVPR'25).
  - Open-Vocabulary 3D Articulated Objects Modeling (Articulate AnyMesh) (arXiv'25).
- **Lab**: Wrap-up and final demos.
- **Final Project**: End-to-end pipeline for 3D per-part segmentation + articulation, evaluated on PartNet-Mobility or Shape2Motion, with optional robotics demo.

Grading

- Homework & labs: 40%
- Final project (report + demo): 50%
- Participation/reading discussions: 10%